

Project Report: T20 World Cup 2022 – Data-Driven Best XI Selection

1. Executive Overview

This report documents the end-to-end analytics workflow used to build a data-driven player selection system for the T20 World Cup 2022. The project combines web scraping, data cleaning, dimensional modeling, and interactive Power BI dashboarding to move cricket team selection from subjective judgment to a criteria-based, evidence-backed process.

The final deliverable is an interactive Power BI dashboard that lets a selector filter 200+ players against role-specific performance thresholds — across five player categories (Power Hitters, Anchors, Finishers, All-Rounders, Specialist Fast Bowlers) — and compare combinations before finalizing a Best XI.

2. Business Problem

Cricket team selection is frequently influenced by reputation, recent media narrative, or subjective "gut feel" rather than a consistent statistical standard applied equally to every player. This creates two risks:

- **Under-selection risk:** In-form players who don't have a big reputation get overlooked.
- **Over-selection risk:** Well-known players get picked despite not meeting the statistical bar for the specific role they're being selected for (e.g., a finisher with too low a strike rate, or a fast bowler with an economy rate too high for the powerplay/death overs).

There was no single, reusable framework to evaluate every player against the *same* role-specific criteria across an entire tournament.

3. Objective

Build an analytics pipeline and dashboard that:

1. Collects complete T20 World Cup 2022 match, batting, bowling, and player data from ESPNcricinfo.
2. Cleans and transforms the data into an analysis-ready format.
3. Models the data into a star schema suitable for fast, flexible filtering in Power BI.
4. Defines explicit, quantified selection criteria for five player roles.
5. Delivers an interactive dashboard where a selector can filter, compare combined statistics of player pairs/groups, and lock in a data-supported Best XI.

4. Dataset

Table	Description	Approx. Size
Match Summary	Team 1, Team 2, winner, margin, ground, date, scorecard link	45 matches
Batting Summary	Per-player, per-match batting figures (runs, balls faced, 4s, 6s, dismissal, position)	~500+ innings records
Bowling Summary	Per-player, per-match bowling figures (overs, runs, wickets, economy)	~450+ bowling spells
Player Info	Player name, team, batting style, bowling style, role, description, image	200+ players

Source: ESPNcricinfo, T20 World Cup 2022 tournament pages. **Collection method:** Bright Data Data Collector (JavaScript-based interaction + parser scripts) — four separate collectors built for match results, batting summary, bowling summary, and player info. Output delivered as JSON.

5. Methodology

5.1 Data Collection (Web Scraping)

- Built four Bright Data collectors targeting: match results table, batting scorecards, bowling scorecards, and individual player profile pages.
- Each collector's interaction script navigated to the target page(s); the parser script extracted structured fields (e.g., `enginetable > tbody > tr.data1` for match results).
- Batting and bowling collectors used a two-stage crawl: Stage 1 collected all match scorecard links from the results page; Stage 2 visited each link and extracted the batting/bowling tables.
- Data delivered as JSON files, one per data type.

5.2 Data Cleaning & Transformation (Python / Pandas)

- Loaded raw JSON files (`match_summary`, `batting_summary`, `bowling_summary`, `player_info`) using Python's `json` module.
- **Match Summary:** Renamed `scorecard` → `match_id` to serve as the primary key linking all other tables.
- **Batting Summary:**
 - Flattened nested per-match batting arrays into a single list using `extend()`, then converted to a DataFrame.

- Engineered an `Out/Not Out` column from the `dismissal` field (blank = Not Out, non-blank = Out) and dropped the raw `dismissal` column.
- Cleaned player names by stripping special/encoding artifacts.
- Solved the team-order ambiguity problem (e.g., "Namibia vs Sri Lanka" vs. "Sri Lanka vs Namibia") by building a lookup dictionary keyed on both team-order permutations, mapped back to `match_id` via `.iterrows()` and `.map()`.
- Exported the cleaned table to CSV.
- **Bowling Summary** and **Player Info** followed a parallel cleaning process (not detailed step-by-step here; full code is in the accompanying notebook).

5.3 Data Modeling (Power BI / Power Query)

- Imported all cleaned CSVs into Power BI via folder connector.
- **dim_players:** Applied "Use First Row as Headers," extracted player name text before the `(C)` captain marker using "Text Before Delimiter," trimmed whitespace, sorted and removed duplicate rows (219 → 213 rows).
- **dim_match_summary:** Added a calculated `Stage` column (Qualifier vs. Super 12) using a conditional column keyed on match date (before 22 Oct 2022 = Qualifier).
- **fact_bowling_summary:** Renamed columns for clarity (`team`, `0s`, `4s`, `6s`), split the `overs` column by delimiter `(.)` into whole overs and partial balls, replaced nulls with 0, and derived a `balls` column (`overs × 6 + partial balls`) to enable accurate strike rate/economy calculations.
- **fact_batting_summary:** Converted the text-based dismissal indicator into a binary `Out` column (0 = Not Out, 1 = Out), and cleaned the captain marker from player names.
- **Relationships (Star Schema):** `fact_batting_summary` and `fact_bowling_summary` linked to `dim_match_summary` via `match_id`; both fact tables linked to `dim_players` via player name (batsman name / bowler name → name). All relationships are one-to-many from dimension to fact.

5.4 DAX Measures & Calculated Columns

Key measures built under organized folders (Batting, Bowling, etc.), including:

- `Total Runs` = `SUM(fact_batting_summary[runs])`
- `Total Innings Batted` = `COUNT(fact_batting_summary[match_id])`
- `Total Innings Dismissed` = `SUM(fact_batting_summary[out])`
- Batting average, strike rate, boundary%, bowling average, bowling strike rate, economy, and dot-ball% measures (full list maintained in the measures reference file)

Calculated columns included:

- `Boundary Runs` = `(fours × 4) + (sixes × 6)`

5.5 Dashboard Build

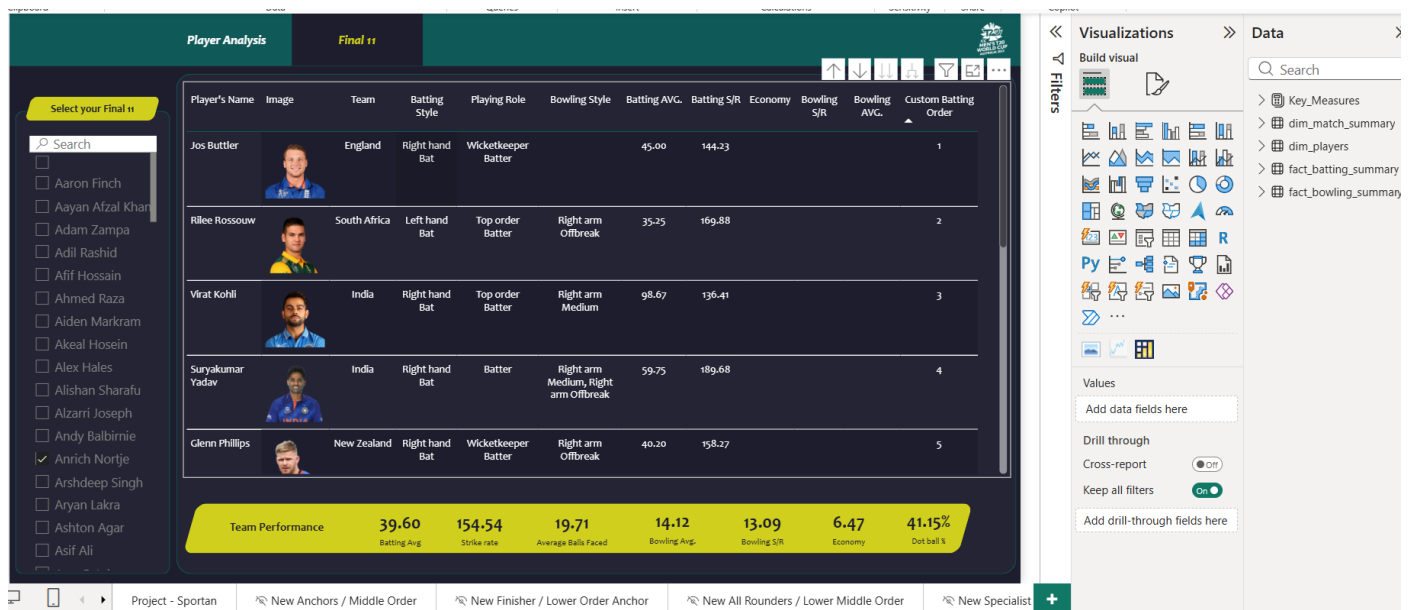
Five role-based analysis pages plus a Final XI comparison page:

Page	Selection Criteria Applied
Power Hitters / Openers	Batting average > 30, Strike rate > 140, Innings batted > 3, Boundary% > 50%, Batting position < 4
Anchors / Middle Order	Batting average and strike rate thresholds for stabilizing innings
Finishers	Higher strike-rate weighting for death-overs hitting ability
All-Rounders	Batting strike rate > 130-140 combined with bowling economy/strike rate thresholds
Specialist Fast Bowlers	Bowling economy < 7, Bowling strike rate < 16, Dot-ball % > 40%

Each page includes a sortable data table, trend line charts (batting average, balls faced, strike rate, boundary%), and a scatter plot (strike rate vs. batting average) to visually separate high performers. A player-selection feature lets the user click two or more players to see their **combined/partnership statistics**, supporting side-by-side "what-if" comparisons.

5.6 Dashboard Preview

Power Hitters / Openers Page — this is the page referenced throughout Section 5.5, showing the shortlist table (Alex Hales, Jos Buttler, Kusal Mendis, Quinton de Kock, Rilee Rossouw), consistency trend lines (batting average, balls faced, strike rate, boundary%), and the Strike Rate vs. Batting Average scatter plot used to visually separate top performers.



Visualizations

Build visual

Filters

Values

Drill through

Cross-report

Keep all filters

Add drill-through fields here

Data

Search

Key Measures

dim_match_summary

dim_players

fact_batting_summary

fact_bowling_summary

6. Key Insights

- Filtering to Batting Avg > 30, Strike Rate > 140, Boundary% > 50%, and Batting Position < 4 narrowed the opener pool to **5 realistic candidates**: Jos Buttler, Alex Hales, Kusal Mendis, Quinton de Kock, and Rilee Rossouw.
- **Jos Buttler** posted the highest batting average among openers (45.00) at a strike rate of 144.23 across 6 innings; **Rilee Rossouw** posted the highest strike rate (169.88) among the shortlist.
- The projected **Buttler–Rossouw opening partnership** averages ~40 runs at 150+ combined strike rate — enough, on average, to clear the team's 180-run target within the first 4 overs (avg. balls faced ~23.9 per pairing sample), with ~60% of their combined runs coming via boundaries.
- **Virat Kohli** and **Suryakumar Yadav** were the clear middle-order anchors — Suryakumar Yadav's strike rate (~190) and average (~60) were the best in the entire eligible pool for that role.
- Fast-bowler filtering (economy < 7, strike rate < 16, dot-ball % > 40%) isolated **Sam Curran, Anrich Nortje, and Shaheen Shah Afridi** as the core attack. Combined, this trio takes a wicket roughly every 11–12 balls at a ~6.0 economy — projecting **~3 wickets inside the first 6 (powerplay) overs** and only ~36 runs conceded in that span.
- In the all-rounder slot, swapping **Hardik Pandya for Marcus Stoinis** improved projected team batting average from 37.76 to 39.6 and strike rate from 151 to 154.4 — illustrating the dashboard's ability to quantify the trade-off of each selection decision.
- The model **correctly excluded Mitchell Starc** despite his reputation: his economy (8.5), bowling average (34), and bowling strike rate (24) all missed the fast-bowler thresholds, and he had only played 3 of the required 4 qualifying innings — demonstrating that the criteria-based approach avoids reputation bias.

7. Business Recommendations

- **Adopt role-specific statistical thresholds** as the first-pass filter for any shortlisting exercise, rather than relying on reputation or recent headlines.
- **Evaluate batting pairs (not just individuals)** for opening and middle-order slots — combined partnership metrics reveal chemistry that individual stats miss.
- **Treat the 6th batting position (all-rounder slot) as a tunable decision**, not a fixed pick — the dashboard shows that the "best" choice (Stoinis vs. Maxwell vs. Pandya) depends on whether the team needs more batting strike rate or a fifth bowling option.
- **Re-run filters against match-specific context** (e.g., pitch type, opposition) — for example, loosening the fast-bowler thresholds to accommodate a specialist spinner on a turning pitch.
- **Institutionalize the dictionary-based match-ID mapping technique** for any future tournament dataset with inconsistent team-name ordering, to avoid manual join

errors.

8. KPI Impact

KPI	Value Achieved
Final XI Team Batting Average	39.6
Final XI Team Strike Rate	154.4
Top-3 Pace Attack Combined Economy	~6.0 runs/over
Top-3 Pace Attack Combined Bowling Strike Rate	1 wicket / 11-12 balls
Projected Powerplay Wickets (first 6 overs, pace trio)	~3 wickets
Data De-duplication (dim_players)	219 → 213 rows (6 duplicate records removed)

9. Business Impact

The project replaced an ad-hoc, reputation-influenced selection process with a **repeatable, transparent, criteria-based decision framework**. Selectors can now:

- Instantly compare any two or more players' combined performance before committing to a batting order or bowling attack.
- Quantify the exact statistical trade-off of swapping one player for another (e.g., the Stoinis-vs-Pandya batting average/strike rate delta).
- Justify selection decisions with tournament-wide, role-specific evidence rather than a handful of recent matches or public sentiment — directly reducing selection bias risk.

10. Limitations

- Data covers only the T20 World Cup 2022 tournament; player form outside this window (domestic leagues, bilateral series) is not factored in.
- No opposition-strength or pitch-condition data is included, so criteria thresholds are applied uniformly regardless of match context.
- Small sample sizes for some players (e.g., minimum innings-played thresholds) can exclude in-form players who simply played fewer matches.
- Web-scraped data is a point-in-time snapshot; any post-tournament corrections to official scorecards would not be reflected without a re-scrape.

11. Conclusion

This project demonstrates a complete, reproducible analytics workflow — from raw web scraping through cleaned, modeled data to an interactive decision–support dashboard — applied to a real–world (ifplayfully framed) selection problem. The resulting Best XI (Buttler, Rossouw/Hales, Kohli, Suryakumar Yadav, Glenn Phillips, Marcus Stoinis, Sikandar Raza, Shadab Khan, Sam Curran, Anrich Nortje, Shaheen Shah Afridi) is fully traceable to quantified, role–specific statistical criteria rather than subjective judgment, and the dashboard supports ongoing "what–if" refinement as new data or match context becomes available.

Appendix: Final XI (as selected via dashboard)

Position	Player	Team	Role
1	Jos Buttler (wk)	England	Opener / Power Hitter
2	Rilee Rossouw (or Alex Hales)	South Africa / England	Opener / Power Hitter
3	Virat Kohli	India	Anchor
4	Suryakumar Yadav	India	Anchor
5	Glenn Phillips	New Zealand	Anchor
6	Marcus Stoinis	Australia	Batting All–Rounder
7	Sikandar Raza	Zimbabwe	Bowling All–Rounder
8	Shadab Khan	Pakistan	Bowling All–Rounder
9	Sam Curran	England	Specialist Fast Bowler
10	Anrich Nortje	South Africa	Specialist Fast Bowler
11	Shaheen Shah Afridi	Pakistan	Specialist Fast Bowler

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